DEMING PRIZE WINNER 2024

TATA POWER-DDL

Towards a Greener Tomorrow

AI, ML, VR, AR, Robotics & Quantum Tech for Indian Power Utilities

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India’s Power Sector Transformation and Digital Imperatives

- **Renewable Energy Expansion:** India is rapidly expanding renewable energy aiming for 500 GW non-fossil capacity by 2030 to reduce emissions.
- **AI and Machine Learning in Grid:** AI and ML convert smart grid data into predictive insights enhancing demand forecasting and fault detection.
- **VR, AR, and Robotics for Utilities:** VR, AR, and robotics improve workforce training, safety, and inspection efficiency in power utilities.
- **Quantum Technologies for Grid Resilience:** Quantum communication and sensing enhance future grid security and operational resilience under India’s National Quantum Mission.

Peer to Peer – Customer value creation

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PROSUMERS *"Have surplus solar power"*

- Additional Revenue Stream
- Better utilization of RTS
- Market based Pricing technology
- Faster solar payback
- Track sales digitally

P2P Benefits

CONSUMERS *"Want affordable green power"*

- Access to Clean Solar energy
- Competitive energy price
- Reduction in Electricity bills
- Transparent Digital Transaction

Promotion Channels

Electricity Bills

- Bill inserts
- QR Code

Website & Mobile App

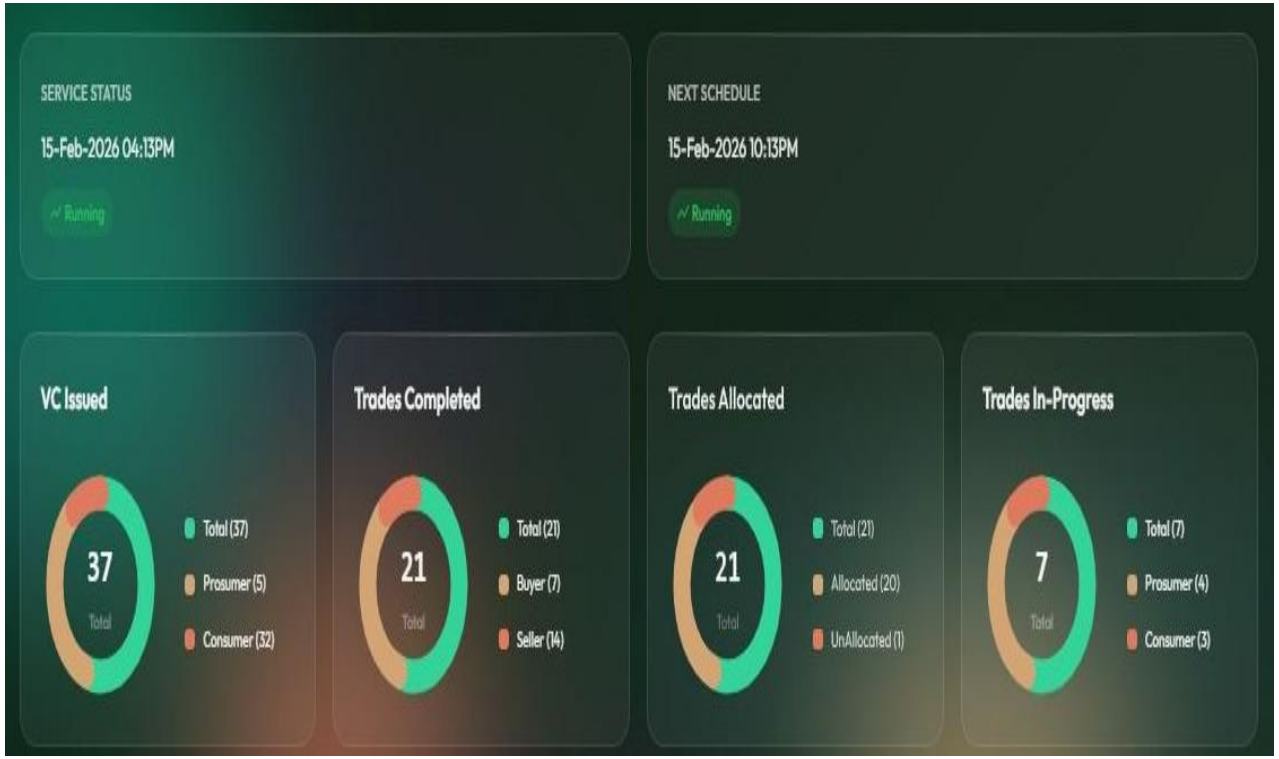
- Web page
- Banners
- Personalized Alerts

SMS, WhatsApp & Email Campaigns

- Targeted to solar consumers
- High end consumers

On-Ground Outreach

- RWAs / IWAs
- Solar Ambassador
- Solar Vendors

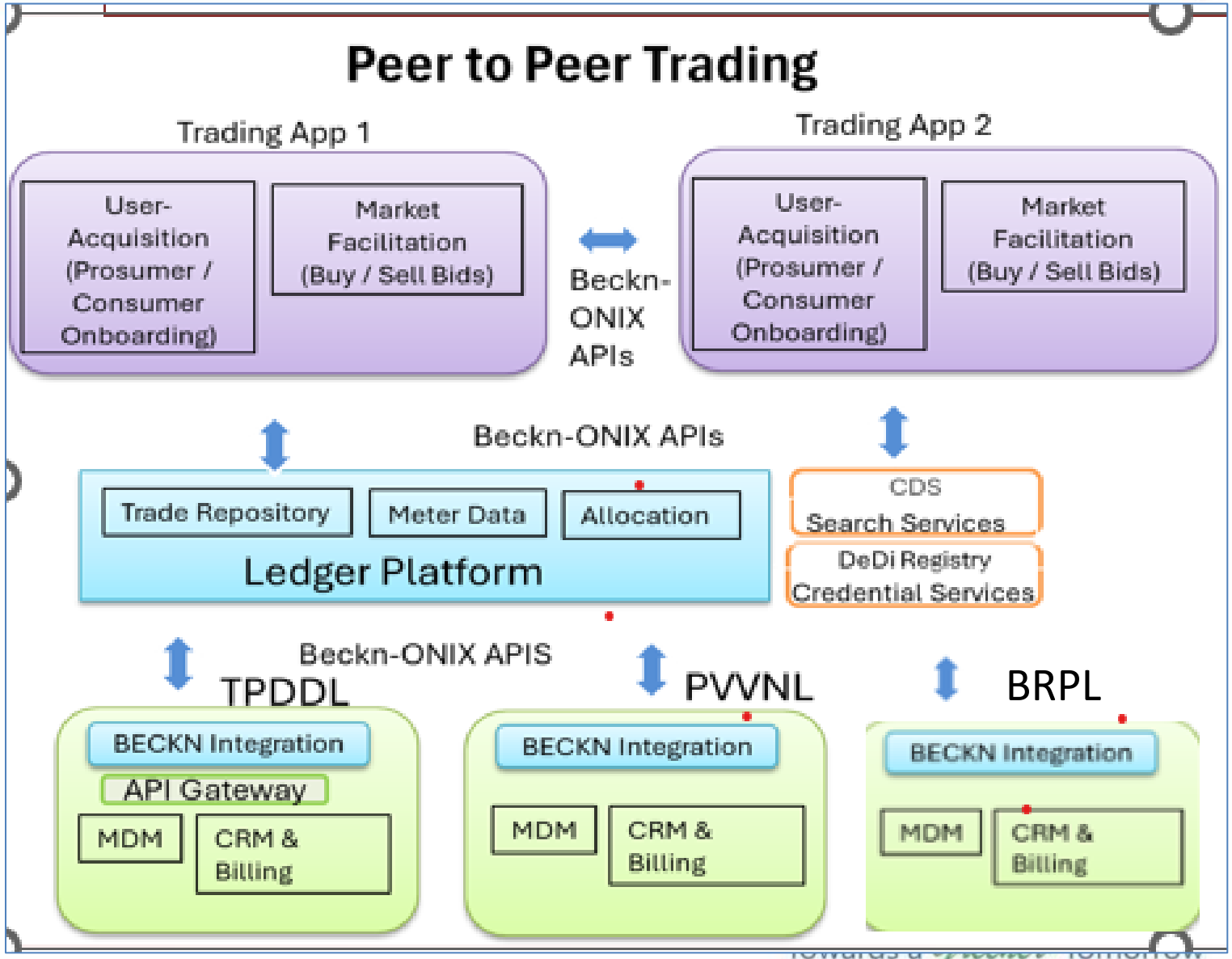


New Technology – Blockchain P2P

- Inter State P2P implemented.
- Need to drive for stability and Scale.

- **Intra Discom (TPDDL) P2P platform**
  Deployed

- ***Interstate (PVVNL, TPDDL, BRPL) P2P Platform***
  Pilot- Ongoing



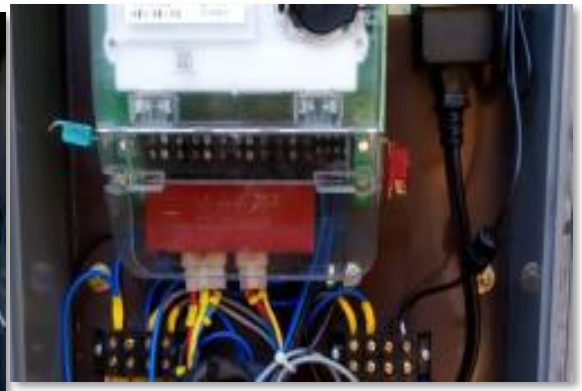
Asset Optimization: Integration of Smart Meters with Distribution Automation & SCADA System

(Application)

In field

Smart Meter redesigned to capture Transformer Sensor Data

a) Meter Redesigned

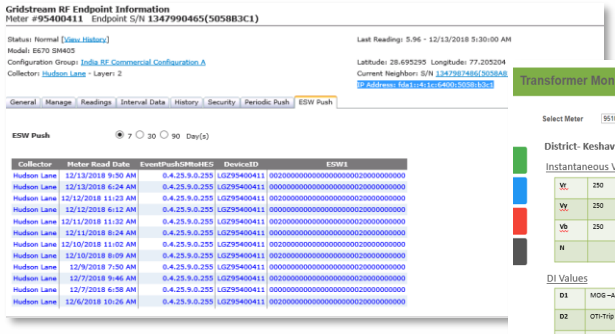


b) Closer look at 5DI and 1 DO

c) Relays activated

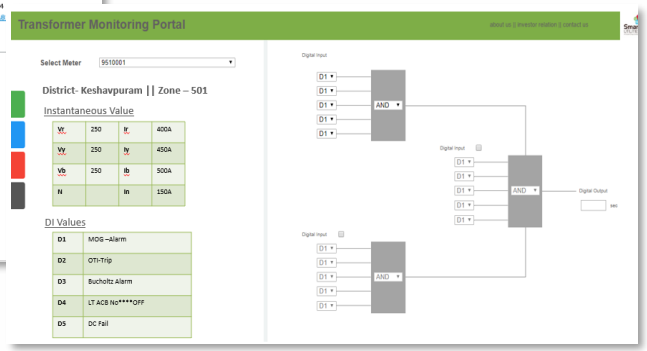
At Backend

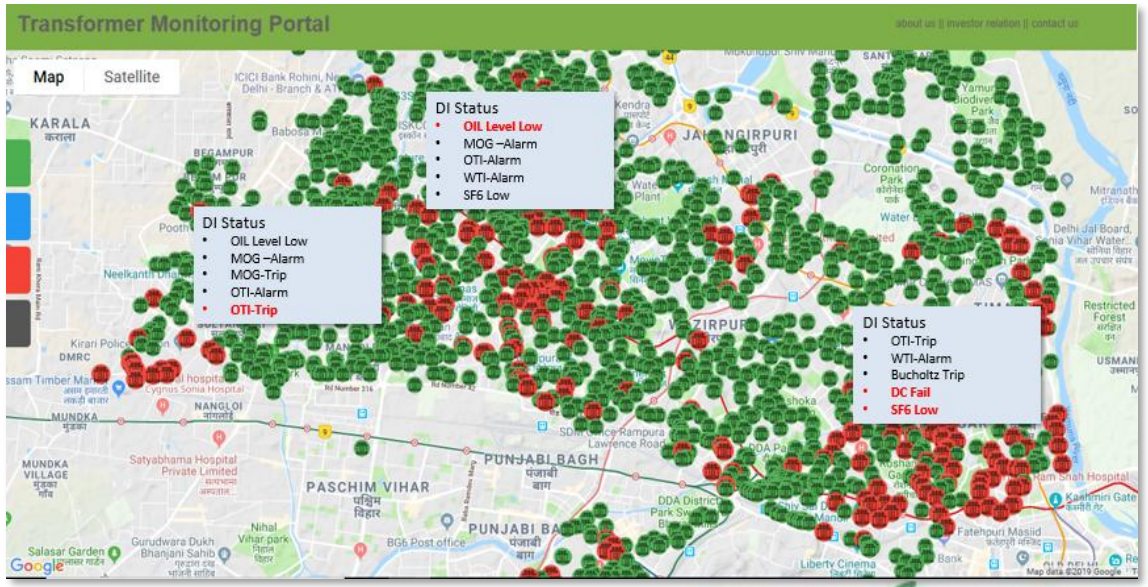
At the Backend Servers (Network Operation Center)



1) Digital Data

2) Algorithms





3) Runtime visual representation to the (24X7) Network Operation Team

Solution : Image Analytics through AI/ML

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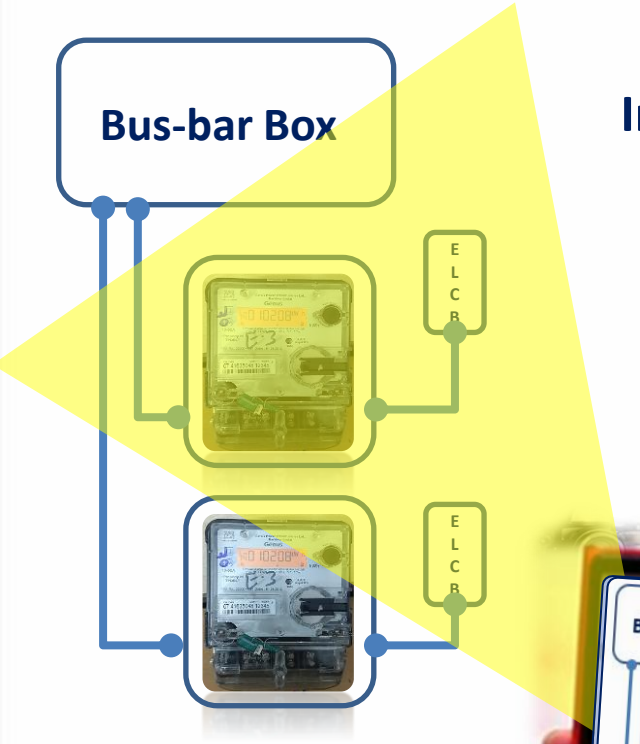
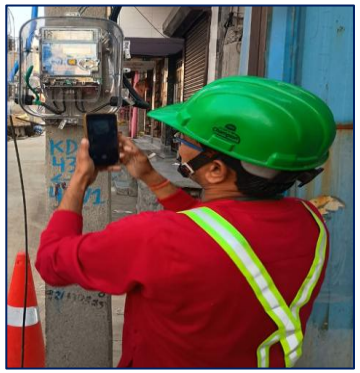


Installation Quality check through AI

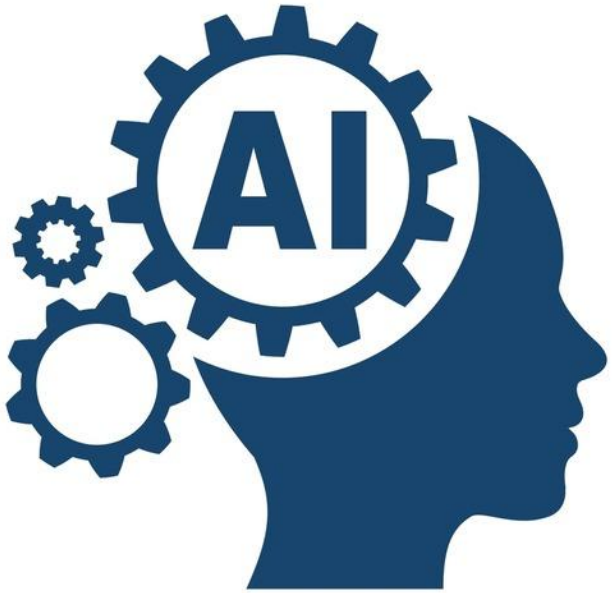
Image Capturing through FFA

Image Analytics through AI

Slot Number	Quality Parameters
1	Meter box Earthing done
2	Meter terminal seal done
3	Meter Box Sticker pasted
4	Meter box cover found sealed
5	No sharp bend in Service cable



AI / ML

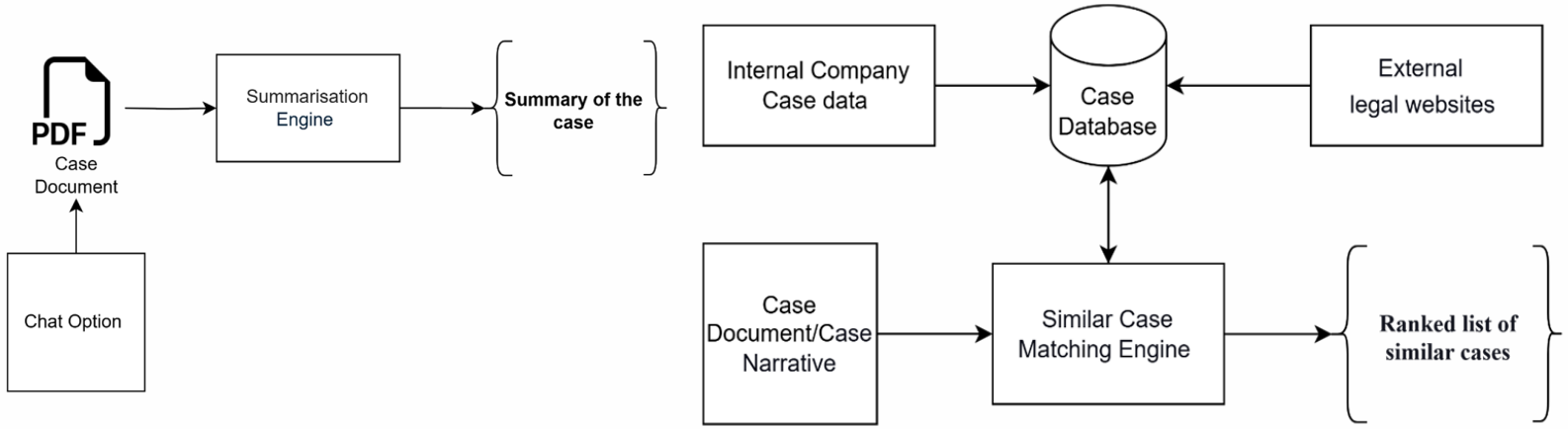


Gen AI Based Legal Assistant

GPTW

Reduction of wasteful expenditure

TATA POWER-DDL



Challenges

- Static Data Dependency
- Manual Process
- Limited Research Capabilities
- Real Time Constraints

Core Features

- Case Summarization
- Interactive Chabot
- Similar Case Discovery

Benefits

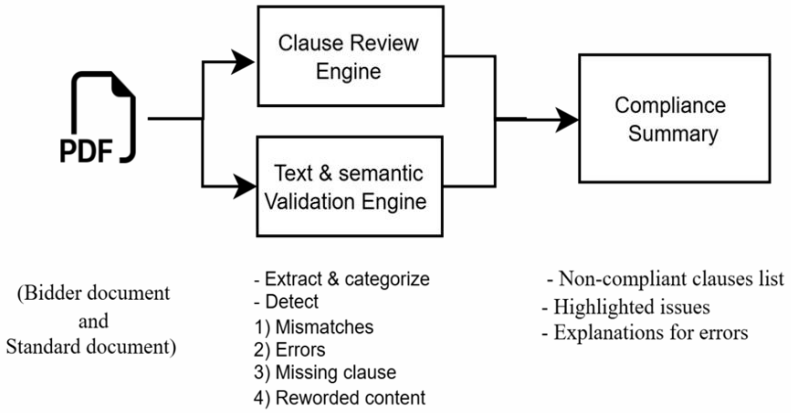
- Quick identification of relevant sections and citations
- Helpful in preparing instant summary of judgements
- Faster review of pleadings.
- Minimalization of efforts
- Suggest legally sound clauses based on the type of contract, jurisdiction, and risks involved.

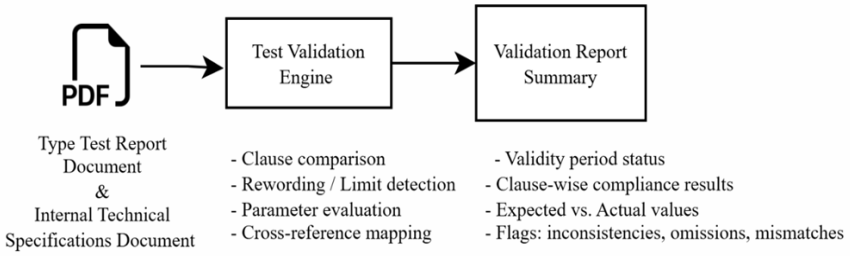
Gen AI Based Tender Evaluation

GPTW

Reduction of wasteful expenditure

TATA POWER-DDL





Challenges

- Technical report evaluation lacks efficiency.
- Manual tender document review is time-consuming
- Complex regulation interpretation requires expertise
- Inconsistent compliance checking across projects

Core Features

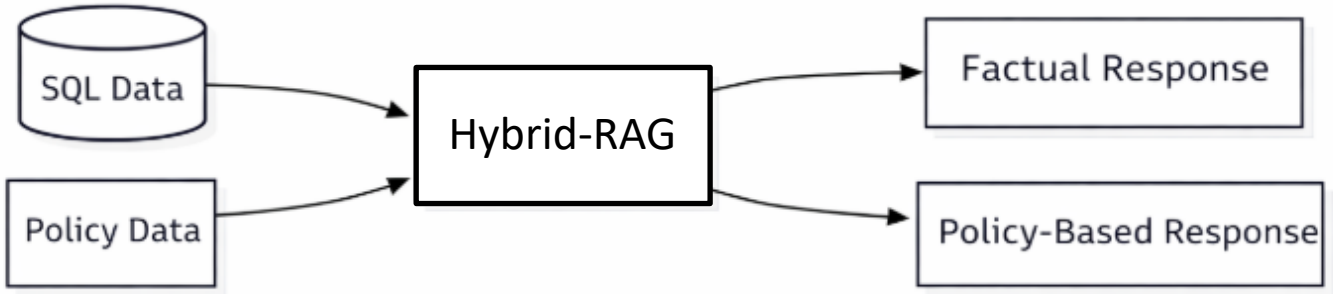
- Document Library
- Compliance Evaluation
- Type Test Validation
- Regulation Q&A & Assistance

Benefits

- Authenticate, compliance checking, compilation and comparison of technical bid documents of all bidders in a tender.
- Ensures accuracy: low to no chances of errors in technical evaluation.
- Enhances productivity of the technical evaluation process

Gen AI Roshni 2.0-Enhancing Customer Experience

GPTW



Challenges

- Difficulties in Accessing Policies
- Inefficiency of Traditional Methods
- Need for personalised solution

Core Features

- Secure Employee Data Access
- Hybrid RAG Technology
- Intelligent Query Handling

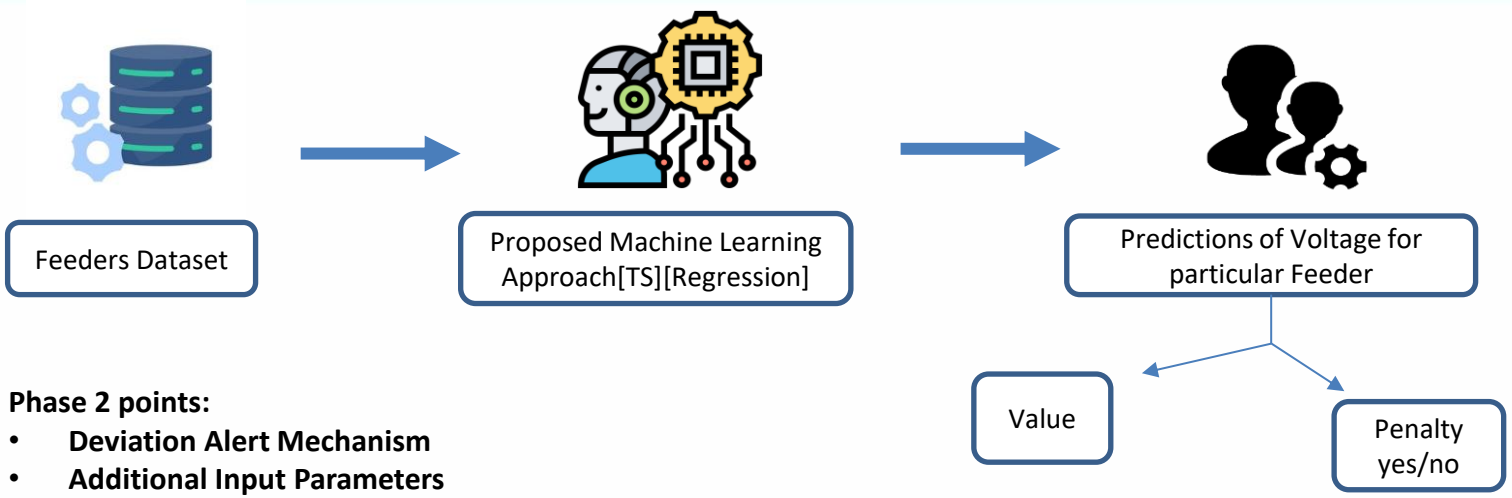
Benefits

- Personalised AI Assistant
- Automation and Efficiency
- Improves employee experience

Voltage Prediction 2.0

Reliability Excellence

TATA POWER-DDL



Phase 2 points:

- **Deviation Alert Mechanism**
- **Additional Input Parameters**
- **Hourly Forecasting**
- **Integration of All Exchange points of TPDDL**

Challenges

- If the voltage falls below **97% (64.02 KV)** or above **103% (67.98 KV)**, a penalty is imposed.
- We need to forecast voltage levels to take preventive actions and minimize penalties

Core Features

- Feeder-wise independent modelling
- End-to-End Automated Data Pipeline
- Use of Hybrid Models
- Hourly future predictions

Benefits

- Feeder Wise hourly insights
- Daily Performance Report
- Better Planning

Human Recourses

GPTW

TATA POWER-DDL

Phase 1

- T&D employee master data for various dimensions like Department, Function, Job family, Date of last transfer, promotion date, retirement date, ethnic origin, grade, Shared service resource flag, designation etc
- Deployed in Delhi, Mumbai and WODL
- Data processing in Data Lake and Visualization in Power BI

TATA POWER

Tata Power Company
T&D - HR Master

0.23% | 9.5%

Select Entity: All | Select Function: All | Select Grade: All | Select Category: All | Select Gender: All | Shared Resource: All | 18730 Emp Count

Discom_Final	Employee Id	Emp Name	Employee Group	Grade	Designation	Department	Function	Job Family
TPC MO & North	171140	Mr Anthony Cajetan Fernandes	Staff	Tech. IV	Chargehand	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	162728	Mr Sanjay Namdeo Garud	Staff	Tech. IV	Chargehand	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	170771	Mr Tribhuwan Singh Kanyal	Staff	Tech. II	Foreman	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	213233	Mr Arnav Ingole	Trainee	ME03	Graduate Engineer Trainee - Electrical	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	205855	Mr Rajshekhar Nagappa Hadpad	Staff	Tech. VI	Technician - B	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	169919	Mr Maheshwar M Swaminathan	Staff	Tech. V	Technician 'A'	Vikhroli-Maintenance	Mumbai Transmission	Operations and Ma
TPC MO & North	213110	Mr Ganesh Pandurang Patil	Trainee	ME03	Graduate Engineer Trainee - Electrical	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	201206	Mr Chetan Mohan Pharande	Management	MD2	Group Head - Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	206952	Mr Kunal Balu Mali	Staff	Tech. X	Junior Works Attendant 1	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	206947	Mr Hemant Nathu Wayal	Staff	Tech. X	Junior Works Attendant 1	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	211595	Mr Dipankar Adhikari	Management	ME03	Lead Engineer - SMC, Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	209492	Mr Mohd Sameer	Management	ME03	Lead Engineer - SMC, Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	207319	Mr Deepak Kumar	Management	ME02	Lead Engineer - SMC, Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	201288	Mr Ravi Mukund Ingle	Management	ME02	Lead Engineer - SMC,Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	200108	Mr Nilesh Pandurang Ghadge	Management	MD2	Lead Engineer-NMC, Vikhroli Node	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	205837	Mr Hemant Pandurang Jadhav	Staff	Tech. VI	Operation Assistant	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	202776	Mr Sajindra Gajanan Jadhav	Staff	Tech. V	Senior Operation Assistant	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPC MO & North	172081	Mr Raghuvir Mahavir Dhanke	Staff	Tech. V	Sr.Operation Assistant	Vikhroli Sub Station	Mumbai Transmission	Operations and Ma
TPSODL	50114	Mr Sadasiba Dash	Consultant	Consultants	Consultant- Security & Vigilance	VIGILANCE & SECURITY	VIGILANCE & SECURITY	Administration, Se
TPSODL	50112	Mr Bipin Bihari Praharaj	Consultant	Consultants	Consultant- Security & Vigilance	VIGILANCE & SECURITY	VIGILANCE & SECURITY	Administration, Se
TPSODL	50079	Mr Somanath Maharana	Consultant	Consultants	Consultant- Security & Vigilance	VIGILANCE & SECURITY	VIGILANCE & SECURITY	Administration, Se
TPSODL	800072	Mr Pradipta Kumar Dash	Consultant	Consultants	Consultant- Security & Vigilance	VIGILANCE & SECURITY	VIGILANCE & SECURITY	Administration, Se
TPSODL	50094	Mr Sudhir Kumar Sahoo	Consultant	Consultants	Consultant- Security & Vigilance	VIGILANCE & SECURITY	VIGILANCE & SECURITY	Administration, Se

Phase 2:

- To develop HR Employee Workforce Dashboard for Tata Power (Mumbai) to visualize key workforce metrics including demographics, diversity, experience, grades, job families, span of control, and retirement projections.
- The dashboard enables data-driven HR insights for effective workforce planning and decision-making.
- Historical data will be available
- To be deployed across all 6 DISCOMS, with DISCOM level access security

Robotics

RPA Use Cases

S.No.	Project Name	Benefits
1	**Deployed in New Connection Process:** -Duplicate Notification -Dues Verification -Batch Allocation	-Approx. 7% rework avoided -Saving: 2 Lacs/year -Saving: 17 Lacs/year -2496 Man-hours optimized
2	**Deployed in Meter Reading** -Reading Quality Check -Temp. Disconnected Billing -Key Customer Billing	-6856 Manhours optimized -Savings approx. 17 lacs/year
3	**Deployed in Billing** -Key Customer Billing & Quality Check -AMI Performance-Auto On Demand Read -Metering Service	-7 no's Manpower optimized -4260 Manhours optimized -AMI Billing Performance
4	**Recovery & Revenue Collection** -Auto Identification of Recovery actions -Revenue Discipline(STOM, DO, Recovery) -Reconnection	-Revenue Enhance 6 Crore per year -Cycle time reduction -3060 Manhours optimized
5	**Power Management & Finance** - Invoices - TDS Compliance	-Saving: 9.2 Lacs/year -4016 Man-hours optimized

Thank You